Calcic Yermosols

Description:
Calcic Yermosols are mineral soils typical of arid and semi-arid regions where evaporation greatly exceeds rainfall.

They are characterized by the accumulation of calcium carbonate ($CaCO_3$) in the upper 50–100 cm, forming a calcic horizon (Bk). These soils are often light-colored (pale brown to grey), coarse- to medium-textured, low in organic matter, and naturally low in fertility.

Limited leaching and high evapotranspiration lead to carbonate concentration and occasional salinity in surface layers.

Drainage:

- **Wet season:** Generally well-drained due to coarse texture, but infiltration can be restricted if a calcrete layer is near the surface.
- **Dry season:** Surfaces become hard and crusted; moisture retention is very low.

Landscape:

- Found on gently undulating plains, alluvial fans, desert basins, and limestone plateaus in arid regions. Typical rainfall ranges between 200–600 mm annually. Common in northern and eastern Kenya (Marsabit, Garissa, Kitui, Makueni) and other semi-arid lowlands.

Depth:

- Moderately deep to deep (50–150 cm); effective rooting may be restricted by hard calcrete or stony layers.

Workability:

- **When wet:** Loose and friable; easy to cultivate.
- **When dry:** Hard, compact, and crusted; tillage and seedbed preparation are difficult without moisture.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–12 cmol(+)/kg).
- **Base saturation:** Very high (>80%) due to abundant calcium carbonate.
- **Nutrients:** Adequate calcium and magnesium; nitrogen, phosphorus, and micronutrients (Zn, Fe) often deficient.
- **Organic matter:** Very low (<1.5%), resulting in weak aggregation and low biological activity.

pH:
Alkaline (7.5–8.5), which reduces the availability of phosphorus and some micronutrients.

Management:

- Incorporate organic matter (manure, compost, green manure) to improve structure and water retention.
- Apply acid-forming fertilizers such as ammonium sulfate to reduce alkalinity slightly.
- Use N and P fertilizers and micronutrient supplements as needed.
- Employ water-harvesting techniques (zai pits, contour bunds) and mulching to conserve moisture.
- Avoid over-irrigation to prevent salinization and maintain drainage.

Suitable Crops:
Drought-tolerant crops like sorghum, millet, cowpeas, pigeon peas, and green grams under rainfed conditions. Under irrigation, crops such as maize, tomatoes, onions, and watermelons can thrive with proper fertility and salinity management.